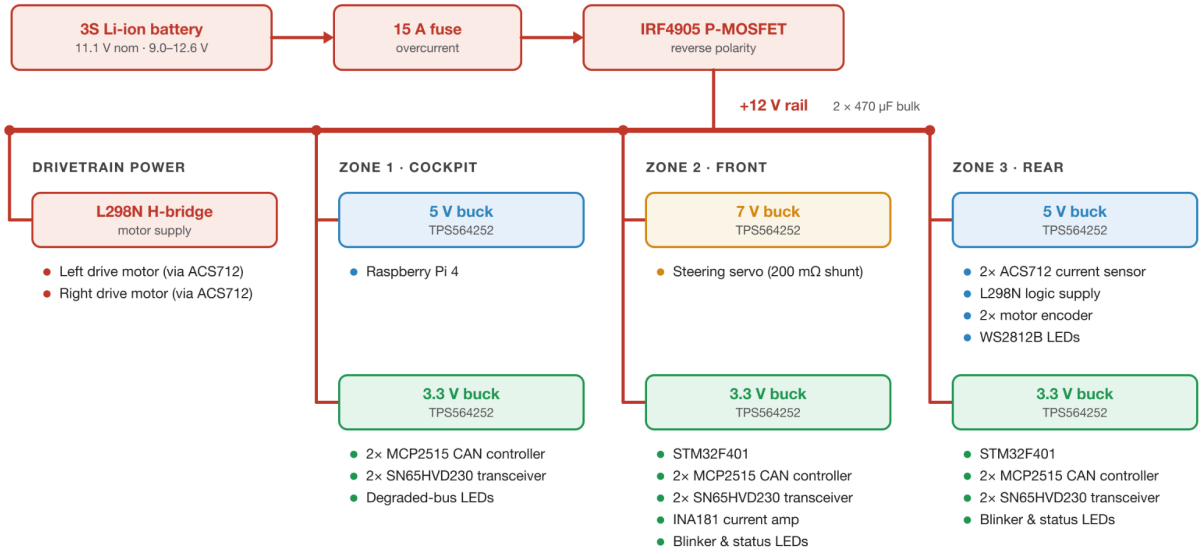
18449 Lab 4 Writeup

Samuel Greco (samuelgr), Nate Miller (nsmiller), Sam Regardie (sregardi)

Design Brief

Power Budget and Battery

- We are using an 11.1V (nominal) 5200mAh lithium battery with an XT60 connector. We are getting a corresponding lithium battery charger and a lipo bag. There will be an XT60 connector soldered to the board which will power our board.
- We will be having independent power for each zone, with only the 12V being shared between the zones. Each zone has independent buck converts to step down to 3.3V, 5V, and 7V as needed

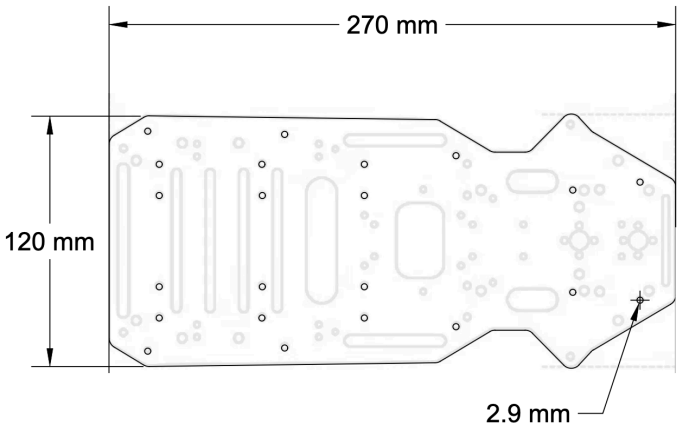


Rail	Source	Peak current	Average current (estimate)	Worst-case voltage
+12V	Battery	About 10 A (handout estimate)	~1A	8.8–12.6 V
+7V_Z2	IC6	3.0 A, servo stall	0.2 A	6.90–7.4 V
+5V_Z1	IC4	3.0 A, Pi 4 supply rating	0.7 A	4.8–5.2 V
+5V_Z3	IC5	0.5 A, motors	0.1 A	4.8–5.2 V

+3.3V_Z2	IC1	100 mA, logic	50 mA	3.2–3.4 V
+3.3V_Z3	IC2	100 mA, logic	50 mA	3.2–3.4 V
+3.3V_Z1	IC3	65 mA, logic	30 mA	3.2–3.4 V

Outline Drawing

We used the 3D model of the car to find accurate hole locations, we exposed what is likely more than necessary on the PCB so that we have flexibility if they are not all in the exact correct locations. We ordered M2.5 screws and standoffs to attach the PCB to the car, and additional standoffs to attach the Pi on top of the PCB.



Part choices

- We chose to use the common parts as much as possible, and tried to use parts from the manufacturers of the common parts when we could, especially for resistors and capacitors.
- We added a fuse holder and fuses we can swap out easily to avoid unnecessary resoldering
- We chose to use the IRF4905 P-type MOSFET, which was recommended for similar projects
- We chose to use the INA181A, a bidirectional current sense amplifier that senses the voltage across our shunt resistor while drawing little power itself
- We chose to put some buzzers and addressable LEDs on the PCB, mainly just for fun

Current Sensing

- We are using ACS712 hall-effect sensors for the current sensing for the rear motors, which are the same we used in Lab 2. They are placed in series with the motor between the H-bridge and motor, and the sensors are taken to a test point
- The steering servo uses a shunt resistor for current sensing instead. The servo only draws a very small amount of current (10s of mA), and the hall-effect sensors see this as just noise. The

shunt resistor is able to effectively measure the current it draws, while not being overloaded by the servo's ~3A stall current

General Design Decisions

- We are using the slope control mode of the CAN transceivers with a 16 MHz clock in order to match to ones we used in class
- We are using a two layer board

Lab 1 Revision

- Things are mostly the same compared to previous labs, but many pin assignments have changed now that we are not limited by the headers on the nucleo boards
- We changed to putting the A and B pins for each encoder on 2 channels of the same timer, so that we can use their hardware encoder mode instead of relying on interrupts for reading the encoders.